Frozen Vision–Language Features Outperform Fine-Tuned CNNs and ViTs for Electronic-Component Recognition on a Leakage-Audited Web-Scraped Dataset

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***Abstract*—Recognising electronic components in internet photographs is a 20-class fine-grained task trained on small, noisy, web-scraped data. We audit a public Kaggle scrape of 3,530 files. It contains 587 redundant exact copies and 13 conflicting-label duplicate groups, and an automated CLIP screen flags 27.8% of the retained images as out of scope. An earlier file-level split leaked duplicates and inflated accuracy by 5.5–13.6 points. Under cluster-grouped 5-fold cross-validation we compare eleven model families, from HOG features and a scratch CNN through ImageNet CNNs and transformers to DINOv2, CLIP and SigLIP, in zero-shot, linear-probe and fine-tuning regimes. A logistic-regression probe on frozen SigLIP So400m features reaches macro-F1 0.824 ± 0.012 with 23k trainable parameters, above every fine-tuned network (ImageNet fine-tunes: 0.548–0.654). Fine-tuning still beats the probe for ImageNet, DINOv2 and CLIP-B backbones, but not for SigLIP. The final calibrated, abstaining model scores 83.7% accuracy (95% CI 79.9–87.0%) on a sealed test split evaluated once, and 82.1% on a separately collected Wikimedia holdout, where its zero-shot head adds 4.2 points. On the earlier protocol's identical test files it scores 87.7%, against the earlier best of 71.3%.**

***Index Terms*—image classification, transfer learning, vision–language models, linear probing, data leakage, label noise, selective classification, electronic components**

I. Introduction

Electronics hobbyists, teaching laboratories and repair benches handle bins of loose parts: resistors, capacitors, relays, crystal oscillators, microcontroller boards. A camera-based recogniser that names the part, or admits that it cannot, is a useful assistant and a natural small-data transfer-learning problem. The classes are visually fine-grained (a diode and a resistor are both small axial cylinders), and the public training data are web scrapes whose class names double as everyday English words (*switch*, *transformer*, *fuse*).

Reported accuracies on such data are hard to interpret. Image-search scrapes contain many copies of the same picture, and a split made at the file level puts copies on both sides [1], [2]. Our own first iteration of this project reported 71–76% accuracy on a file-level split. As shown below, that split leaked, and the leakage also compressed the gaps between model families. This paper therefore rebuilds the evaluation from the data up and asks: (i) how much does each model family and pretraining source contribute when tuning and evaluation are leakage-free; (ii) with $\sim$2.5k training images, should a pretrained backbone be frozen or fine-tuned; (iii) what does accuracy cost; and (iv) how well does a deployable, calibrated model do on images it has never seen?

Contributions.

- A *data audit* of the 20-class Kaggle *electronic-component-recognizer* scrape [3]: exact and near-duplicate clustering, conflicting labels, an out-of-scope screen and a confident-learning label audit. The audit quantifies how much a file-level split inflates accuracy (5.5–13.6 pts).
- A *leakage-free benchmark* of eleven model families under three adaptation regimes (zero-shot, frozen linear probe, fine-tuning). It uses cluster-grouped 5-fold cross-validation, per-fold paired comparisons and a sealed test split that was read once.
- The finding that *pretraining source dominates*. Frozen SigLIP features with a 23k-parameter linear head beat every fine-tuned network, while fine-tuning still helps the weaker ImageNet, DINOv2 and CLIP-B backbones.
- A *calibrated, abstaining model* exported to ONNX and evaluated on two sealed sets: an in-distribution test split and an independently collected Wikimedia Commons holdout. We also evaluate it on the original dataset with a seen/unseen breakdown, and replay the earlier protocol for a like-for-like comparison.
- Ablations reported *with their null results*: learning-rate-schedule fix, test-time flipping, training-data cleaning and zero-shot fusion.

II. Related Work

Transfer learning and architectures. Features learned on ImageNet [4] transfer to small target tasks [5], and architectures that are more accurate on ImageNet tend to transfer better, both as fixed feature extractors and when fine-tuned [6]. We compare residual CNNs [7], [8], EfficientNetV2 [9], MobileNetV3 [10], ConvNeXt [11], the Vision Transformer [12],

[13] and Swin [14], plus a VGG-style CNN [15], [16] trained from scratch and a HOG [17] + SVM [18] baseline.

Self-supervised and vision–language pretraining. Self-distillation (DINO [19], DINOv2 [20]) learns general features without labels. Contrastive image–text pretraining (CLIP [21], ALIGN [22], OpenCLIP [23]) adds zero-shot classification from class-name prompts. SigLIP replaces the softmax contrastive loss with a pairwise sigmoid loss [24], and SigLIP 2 adds captioning and self-distillation objectives [25].

Linear probes versus fine-tuning. Linear probes on frozen features are the standard representation-quality test [20], [21]. Fine-tuning can distort pretrained features and underperform out of distribution [26]. Interpolating fine-tuned and zero-shot weights recovers robustness [27]. Our fused probe-plus-zero-shot head is a simpler, output-space relative of that idea.

Leakage, duplicates and label noise. Near-duplicates between training and test splits inflate benchmark scores [1]. Leakage is a common cause of irreproducible ML results [2], and networks readily memorise training images [28]. Embedding-based near-duplicate removal is also used to curate web-scale training data [29]. Grouped (blocked) cross-validation is the remedy for structured dependence [30]. Selecting a model on the same folds that score it biases the estimate [31], and freshly collected test sets can reveal accuracy drops that the original test set hides [32]. Confident learning finds label errors from out-of-sample probabilities [33], and such errors are pervasive even in test sets [34].

Calibration and selective classification. Modern networks are miscalibrated, and temperature scaling is a strong fix [35]. Rejecting low-confidence inputs trades coverage for accuracy [36], [37]; the maximum softmax probability is the usual confidence score [38].

Electronic-component recognition. Much of the vision work on electronics targets printed-circuit-board inspection under controlled imaging [39]. For classification of loose components, Shao *et al.* compare twelve fine-tuned ImageNet models and report 92.26–99.95% accuracy [40]. Zhou and Zhang propose a CNN for components on diverse backgrounds [41]. Their data and splits differ from ours, so the numbers are not comparable. Our audit shows why the split construction must be reported alongside any such accuracy.

III. Dataset and Audit

Source. The images come from the Kaggle *electronic-component-recognizer* dataset [3], a web scrape with 20 class folders: battery holder, breadboard, capacitor, connector, crystal oscillator, diode, fuse, heat sink, IC chip, inductor coil, jumper wire, LED, microcontroller board, potentiometer, power supply module, relay, resistor, switch, transformer and transistor. The formats are mixed: among the 2,943 unique decoded files there are 1,486 JPEG, 787 WebP, 402 PNG, 245 AVIF, 20 GIF and 3 MPO images, and 345 needed alpha compositing. A build script decodes every file, collapses exact duplicates by SHA-256, links near-duplicates and writes a versioned manifest with one row per unique image (Table I).

TABLE I
Data-cleaning pipeline. Near-duplicate and out-of-scope rows are flags on retained rows, not removals.

Stage	Images
Raw image files	3,530
Unique files (SHA-256)	2,943
Rows kept (conflicting-label exact duplicates removed)	2,930
Duplicate clusters (grouping unit for every split)	2,787
Rows in near-duplicate clusters	234
Rows flagged out-of-scope (automated CLIP screen)	816

Exact duplicates and conflicting labels. The 3,530 raw files contain only 2,943 unique images: 533 duplicate groups hold 587 redundant copies. Thirteen exact-duplicate groups carry two different labels (e.g. the same picture under *fuse* and *relay*). All 13 groups were excluded, leaving 2,930 rows.

Near-duplicates. Perceptual hashing (pHash radius 10) combined with CLIP cosine similarity ($\geq$0.85 with a pHash match, $\geq$0.92 without) links 234 rows into 91 near-duplicate clusters of size 2–29. Two of these clusters mix labels; their 4 rows are kept and flagged. With the singletons there are 2,787 clusters.

Out-of-scope content. A CLIP zero-shot screen scores each image against the 20 class prompts and nine distractor concepts (person, car, food, logo, text screenshot, cartoon/game art, circuit schematic, website screenshot, collage). It flags an image when the distractor probability mass exceeds 0.25. It flags 816 rows (27.8%; 704 dev, 112 test). The most frequent top distractors are circuit schematics (245), logos (180), cartoon/game art (130) and people (83). Contamination concentrates in classes named by common words (Fig. 1). 121 of 177 *switch* images are flagged, mostly Nintendo Switch consoles and game art, as are 64 of 128 *transformer* images (toy robots, cartoons). The screen is automated and was not human-verified; some flags, e.g. resistor schematics, are arguably in scope.

Label errors. A confident-learning-style audit [33] on out-of-fold probe probabilities lists 69 confident label issues, 11 of which are also out-of-scope flags. The most common patterns are diode→transistor, microcontroller board→IC chip and jumper wire→breadboard (4 each).

Decision. Flagged rows were *kept* for training and evaluation. We report all-row scores and *in-scope* scores over rows without the out-of-scope flag. Section VII shows that removing the flagged rows from training hurts.

IV. Evaluation Protocol

Splits. Whole duplicate clusters are assigned to splits (seed 42), so copies and near-copies of an image never sit on both sides of any split [30]. A sealed *test* split of 418 rows (16–25 per class) is held out first. The remaining 2,512 *dev* rows form five cluster-grouped folds. The largest total-variation distance between class distributions across the outer grouped split is 0.017. All model selection used dev folds only. The test split is readable only through an explicit confirmation flag. It was read *once*, for the final model, after every design choice had been

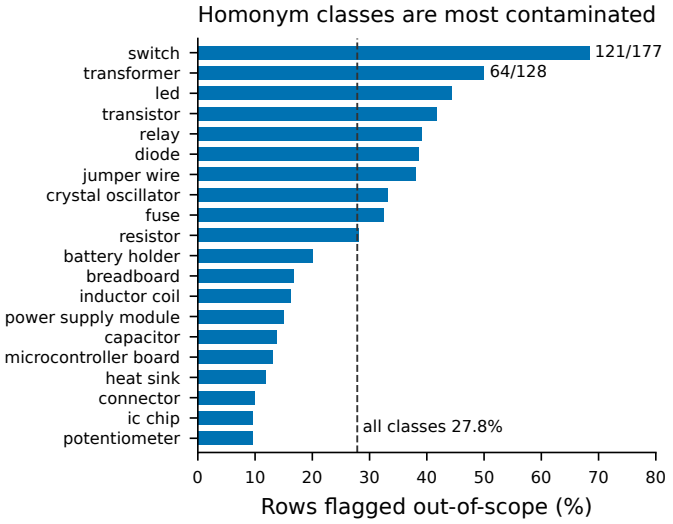


Fig. 1. Share of each class flagged out-of-scope by the automated CLIP distractor screen (threshold 0.25; n = 2,930 rows). Dashed line: all classes. Numbers give flagged/total for the two worst classes.

frozen. A second sealed set, the web holdout (Section IX), was collected after the final model was frozen.

Metrics. The primary metric is macro-F1 over the 20 classes. Accuracy is secondary. Cross-validated scores are mean $\pm$ SD over the five folds. In-scope accuracy is pooled over folds. Test accuracies carry 95% Wilson intervals [42]. With 418 test images the binomial standard error is ± 1.8 points, so test-split differences below about 3.5 points are not resolvable. Calibration is measured by the expected calibration error (ECE, 15 equal-width bins) [35]. Selective prediction is summarised as coverage (fraction answered) and accepted accuracy (accuracy on answered images) [37].

Why the earlier split leaked. The first iteration of the project split *files*, not clusters (a 70/15/15 split). Of the 853 images in its validation and test splits, 198 (23.2%) duplicate a training image. Table II replays three linear probes under both protocols. On the duplicated rows the probes are nearly perfect (95–99%). The legacy-style score is inflated by 5.5 points for SigLIP and by 11.0 and 13.6 points for ConvNeXt-T and ResNet-50: weaker models gain most, because memorised copies stand in for the features they lack. The inflation therefore also *compresses* the gaps between families. Accuracy on the non-duplicated legacy rows is still above grouped-CV accuracy (e.g. 0.860 vs. 0.825 for SigLIP). Near-duplicates missed by the exact-duplicate count and differences between the evaluation sets are the likely causes; we do not decompose this residual further.

V. Models and Adaptation Regimes

Table III lists the families, grouped by *pretraining source*, which turns out to be the main axis of variation. Pretrained weights come from `timm` [8] and `open_clip` [23].

Zero-shot (vision–language models only): each image is assigned to the class whose template-averaged text embedding ("a photo of a {class}.", ...) has the highest cosine similarity.

TABLE II
Leakage simulation: accuracy of linear probes under grouped CV and under the legacy file-level split (853 evaluation rows, 198 of them duplicates of a training image).

Probe backbone	Grouped CV	Legacy split	Infl. (pts)	Dup. rows	Other rows
ResNet-50	0.5215	0.6577	+13.6	0.9949	0.5557
ConvNeXt-T	0.5983	0.7081	+11.0	0.9646	0.6305
SigLIP So400m	0.8252	0.8804	+5.5	0.9495	0.8595

TABLE III
Model families. IN = ImageNet. Parameter counts are the trainable parameters of the fine-tuning runs; (B): ViT-B variant; SigLIP So400m has 428.2M.

Family (variants)	Pretraining	Params (M)
HOG + HSV hist. + SVM [17], [18]	none (hand-crafted)	–
VGG-style CNN [15], [16]	none (from scratch)	1.2
ResNet-50 [7], [8]	IN-1k	23.5
EfficientNetV2-S [9]	IN-21k→1k	20.2
MobileNetV3-L [10]	IN-1k	4.2
ConvNeXt-T [11]	IN-22k→1k	27.8
ViT-B/16 [12], [13]	IN-21k→1k	85.8
Swin-T [14]	IN-22k→1k	27.5
DINOv2 ViT-S/B/L-14 [20]	LVD-142M, self-distill.	85.7 (B)
CLIP ViT-B/16, L/14(@336) [21]	400M image–text pairs	85.8 (B)
SigLIP, SigLIP 2 B/16, So400m [24], [25]	WebLI image–text	92.9 (B)

No training images are used. **Linear probe**: the backbone is frozen. Features of the evaluation-preprocessed image feed a standardised logistic regression whose C is chosen inside each training fold by grouped CV. The head has $20(D+1)$ parameters, about 23k for $D = 1152$. **Fine-tune**: the pretrained network gets a new head and is trained with one hand-set recipe per family: AdamW [43], linear warm-up then cosine decay [44], label smoothing 0.1 [45], RandAugment [46] and fp16 mixed precision. Where configured, the recipe adds layer-wise learning-rate decay and frozen-backbone warm-up epochs. Early stopping uses fold-validation macro-F1. SigLIP So400m (428.2M parameters) is too large to fine-tune fully on an 8 GB GPU, so only its last four blocks are trained (*partial fine-tune*, 76.2M trainable parameters, folds 0–2 only). **Classical**: HOG + HSV colour-histogram features with an RBF-SVM. All training ran on one NVIDIA RTX 4060 Ti (8 GB).

VI. Results

A. Main benchmark

Table IV lists the headline rows and Fig. 2 shows every backbone and regime.

Web-scale vision–language pretraining dominates. The best entries are all SigLIP / SigLIP 2 So400m variants. Zero-shot SigLIP So400m (0.776), which uses no training images, beats every fine-tuned network except the partial fine-tune of the same backbone. CLIP ViT-L/14 probes reach 0.715–0.721. Self-supervised DINOv2 is clearly weaker as a frozen feature (ViT-L 0.667, ViT-B 0.633). Among ImageNet-pretrained networks the transformers (ViT-B 0.654, Swin-T 0.639) and ConvNeXt-T

TABLE IV
HEADLINE RESULTS, GROUPED 5-FOLD CV OVER 2,512 DEV IMAGES (MEAN ± SD OVER FOLDS). ZS = ZERO-SHOT, LP = LINEAR PROBE, FT = FINE-TUNE. †PARTIAL FINE-TUNE, FOLDS 0–2.

Backbone	Reg.	Macro-F1	Acc.	In-scope
SigLIP So400m/14@384	LP	**0.824** ± 0.012	0.825	0.861
SigLIP 2 So400m/14@384	LP	0.814 ± 0.011	0.815	0.852
SigLIP So400m/14@384†	FT	0.812 ± 0.023	0.813	0.835
SigLIP So400m/14@384	ZS	0.776 ± 0.018	0.775	0.827
SigLIP 2 B/16	LP	0.758 ± 0.014	0.760	0.792
SigLIP 2 B/16	FT	0.733 ± 0.027	0.735	0.767
CLIP ViT-L/14@336	LP	0.721 ± 0.019	0.727	0.753
CLIP ViT-B/16	FT	0.702 ± 0.017	0.707	0.724
DINOv2 ViT-B/14	FT	0.668 ± 0.024	0.672	0.717
DINOv2 ViT-L/14	LP	0.667 ± 0.020	0.668	0.721
ViT-B/16 (IN-21k)	FT	0.654 ± 0.032	0.659	0.712
ConvNeXt-T	FT	0.641 ± 0.034	0.647	0.689
Swin-T	FT	0.639 ± 0.034	0.644	0.694
EfficientNetV2-S	FT	0.608 ± 0.024	0.616	0.653
ResNet-50	FT	0.580 ± 0.012	0.585	0.618
MobileNetV3-L	FT	0.548 ± 0.016	0.556	0.584
Scratch CNN	FT	0.355 ± 0.015	0.365	0.379
HOG + HSV + SVM	–	0.206 ± 0.017	0.220	–

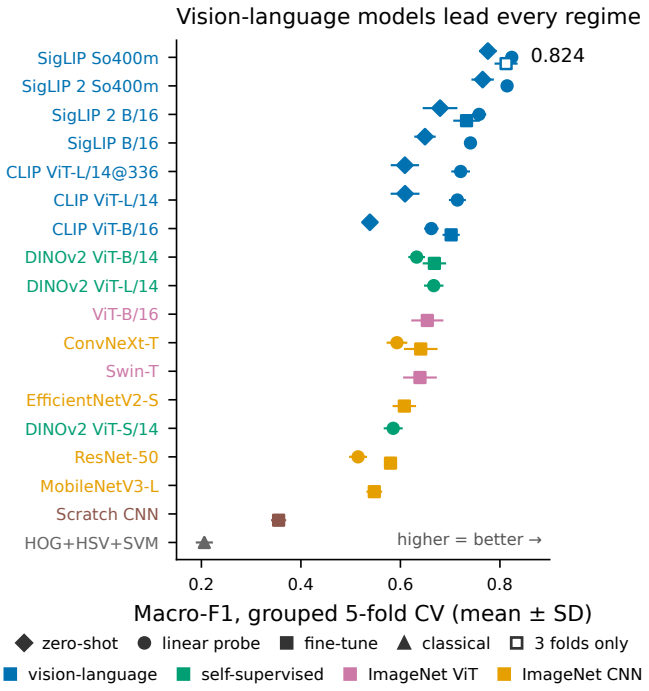


Fig. 2. Macro-F1 of every backbone under each adaptation regime (grouped 5-fold CV, n = 2,512 dev images; bars: SD over folds). Label colour encodes the pretraining source. The open square (SigLIP So400m partial fine-tune) was run on folds 0–2 only. All fine-tunes use the fixed learning-rate schedule.

(0.641) lead; ResNet-50 (0.580) and MobileNetV3-L (0.548) trail. Pretraining *data* matters more than architecture. Four ViT-B models with 86–93M parameters span 8 points: ViT-B/16 with ImageNet-21k weights (0.654), DINOv2 ViT-B/14 (0.668), CLIP ViT-B/16 (0.702) and SigLIP 2 B/16 (0.733), all fine-tuned.

Learned features are indispensable. The scratch CNN (0.355) and HOG + SVM (0.206) are far below every pretrained model. The 19–30-point gap from the scratch CNN to the ImageNet fine-tunes measures what pretraining buys at this data size.

TABLE V
PAIRED PROBE VS. FINE-TUNE MACRO-F1 ON IDENTICAL FOLDS. Δ = FINE-TUNE − PROBE.

Backbone	Folds	Probe	Fine-tune	Δ (pts)	FT wins
SigLIP 2 B/16 (full)	0–4	0.758	0.733	−2.52	0/5
SigLIP So400m (partial)	0–2	0.822	0.812	−0.92	1/3
DINOv2 ViT-B/14	0–4	0.633	0.668	+3.57	5/5
CLIP ViT-B/16	0–4	0.662	0.702	+3.99	5/5
ConvNeXt-T	0–4	0.593	0.641	+4.78	5/5
ResNet-50	0–4	0.515	0.580	+6.52	5/5

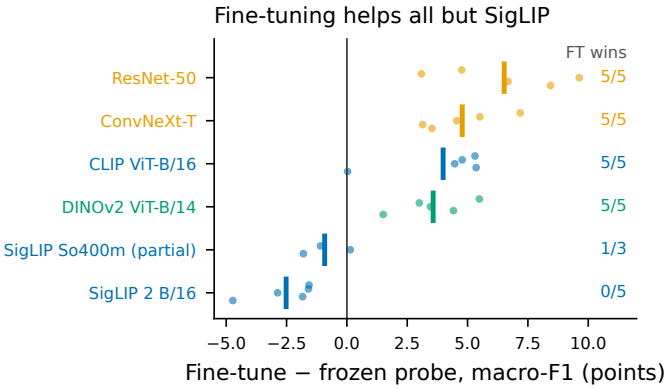


Fig. 3. Per-fold macro-F1 gain of fine-tuning over the frozen linear probe of the same backbone (dots: folds; bars: means; right: folds where fine-tuning wins). SigLIP So400m uses the partial fine-tune on folds 0–2.

Out-of-scope rows cost every family. In-scope accuracy exceeds all-row accuracy for every method, by 1.3–5.7 points (0.861 vs. 0.825 for the best probe). Part of the remaining error comes from the contaminated data rather than from the classifier.

B. Frozen probe versus fine-tuning

Table V and Fig. 3 pair each fine-tune with the frozen probe of the same backbone on identical folds. The answer depends on the backbone. Fine-tuning SigLIP 2 B/16 *loses* 2.5 points to its own probe (0/5 folds better). The SigLIP So400m partial fine-tune loses 0.9 points (1/3 folds), within fold noise. For CLIP ViT-B/16, DINOv2 ViT-B/14, ConvNeXt-T and ResNet-50, fine-tuning gains 3.6–6.5 points and wins on 5/5 folds. The "frozen ≥ fine-tuned" result therefore holds for SigLIP only. Even the best fine-tune of any family (SigLIP So400m partial, 0.812) does not beat the frozen probe of the same backbone (0.822 on the same folds). This pattern is consistent with reports that fine-tuning can distort strong pretrained features [26]. One caveat runs against the probes: fine-tunes early-stop on the fold they are scored on, which biases their CV scores upward [31]. The SigLIP result is therefore conservative.

C. Accuracy versus cost

Among fine-tunes, accuracy follows the pretraining source rather than compute. MobileNetV3-L trains in 0.7 min per fold

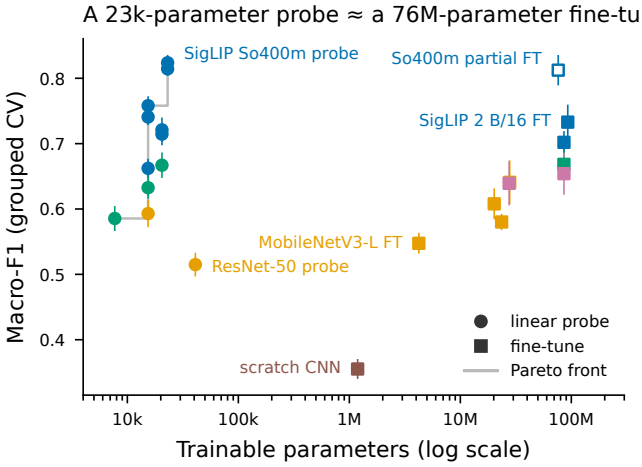


Fig. 4. Macro-F1 vs. trainable parameters for fine-tunes and linear probes (probe = $20(D+1)$ weights; zero-shot models have none and are omitted). Colours as in Fig. 2; bars: SD over folds; open square: 3 folds. Grey: Pareto front.

TABLE VI
ABLATIONS ON THE SIGLIP SO400M MODEL (MF1 = MACRO-F1, ACC = ACCURACY, IS = IN-SCOPE ACCURACY).

Ablation	Metric	Base	Variant	Δ (pts)
hflip TTA, SigLIP So400m probe	MF1	0.824	0.819	−0.50
hflip TTA, SigLIP 2 So400m probe	MF1	0.814	0.820	+0.57
drop 69 label issues (train)	IS	0.861	0.863	+0.16
drop OOD-flagged rows (train)	IS	0.861	0.852	−0.89
zero-shot fusion, dev OOF	Acc	0.825	0.826	+0.16
zero-shot fusion, sealed test	Acc	0.845	0.837	−0.72
zero-shot fusion, web holdout	Acc	0.779	0.821	+4.18
partial FT, 137M, old schedule	MF1	0.822	0.821	−0.10
partial FT, 76M, fixed schedule	MF1	0.822	0.812	−0.92

and ResNet-50 in 1.6 min. The SigLIP So400m partial fine-tune needs 17.5 min (76.2M trainable parameters, 3.8 GB peak VRAM) and still does not beat its probe. On the trainable-parameter axis the Pareto front consists almost entirely of linear probes (Fig. 4). A 23k-parameter logistic regression on frozen SigLIP So400m features matches or exceeds a 76M-parameter fine-tune. The relevant deployment cost is therefore the inference cost of the frozen backbone (Section VIII), not training.

VII. Ablations

Table VI summarises the ablations. Most of the effects lie inside the noise floor of this data size.

Partial fine-tuning. Two partial fine-tunes of SigLIP So400m were run on folds 0–2: 137.2M trainable parameters with the old schedule (0.821, 23.3 min/fold) and the last four blocks, 76.2M parameters, with the fixed schedule (0.812, 17.5 min/fold). The probe scores 0.822 on the same folds. Neither beats it, so the final model keeps the backbone frozen.

Learning-rate schedule fix (null result). The first fine-tuning runs did not restart the warm-up when the backbone unfroze. The fix also froze BatchNorm statistics during the head-only phase. Evaluated on folds 0–2 for ten families, it improved 14 of 30 fold scores, with a mean change of +0.19 points (exact two-sided sign test $p = 0.86$). The SD of the per-fold differences is 1.4 points, a ± 1.6-point noise floor (2 SE) for a 3-fold mean. Only the BatchNorm CNNs moved consistently upward: ResNet-50 (+2.0, 3/3 folds), EfficientNetV2-S (+1.6, 2/3) and MobileNetV3-L (+0.8, 3/3). This points to the BatchNorm change rather than the warm-up, but the two are confounded.

Test-time flipping changed probe macro-F1 by −0.50 (SigLIP, 1/5 folds better) and +0.57 points (SigLIP 2, 5/5 folds better). The effect is inconsistent and below one point, so TTA, which doubles inference cost, was not adopted.

Training-data cleaning hurts. Dropping the 69 audited label issues from training changed in-scope accuracy by +0.16 points, which is noise. Dropping all out-of-scope-flagged rows *reduced* in-scope accuracy by 0.9 points and all-row accuracy by 3.3 points (0.825→0.792). Many flagged rows evidently still carry class signal, or resemble the flagged evaluation images. The training data were left unchanged.

Zero-shot fusion. The final model adds the backbone's zero-shot log-probabilities to the probe's, $\log p_{\text{probe}} + 0.25 \log \text{softmax}(50 \cos)$, with weight and temperature fixed a priori. In distribution this is a null result: +0.16 points out-of-fold and −0.72 points on the sealed test, a tie within the ± 1.8-point standard error. Per-image probe-only test predictions were not saved, so no paired (McNemar) test is possible. Fusion was kept because it is free at inference (one extra 1152×20 product). On the out-of-source web holdout it adds 4.2 points (Section IX).

VIII. Final Model and Deployment

Model. The final model is the frozen SigLIP So400m/14 image tower at 384 px. Its 1152-d feature feeds (i) a standardised logistic-regression probe ($C = 0.01$) refit on all 2,512 dev rows and (ii) the zero-shot text head, fused as above. Preprocessing resizes the short side to 427 (bicubic), centre-crops to 384 and normalises with mean = std = 0.5. A temperature $T = 0.8977$, fit on dev out-of-fold fused scores, lowers out-of-fold ECE from 0.050 to 0.029 [35]. The model abstains when its maximum probability is below 0.5. This is the smallest grid threshold whose out-of-fold accepted accuracy is at least 0.90 (coverage 0.846 at 0.909 accepted accuracy).

Sealed test (evaluated once). The model scores accuracy 0.837 (95% CI 0.799–0.870) and macro-F1 0.834 on the 418 sealed test images (Table VII), within one standard error of its dev out-of-fold estimate (0.826). There is thus no sign of selection overfitting. In-scope accuracy is 0.869 ($n = 306$); the 112 out-of-scope-flagged rows score 0.750. Errors are spread over many class pairs, and no off-diagonal confusion cell exceeds 2 images. The clear weakness is *resistor* recall: 0.545 (12/22, precision 1.00), with the ten misses scattered over seven classes. Power supply module (F1 0.711) and capacitor (0.788) come next. The high *switch* score (recall 0.96) is partly an artefact. 19 of the 25 test switch images are out-of-scope flagged (mostly Nintendo Switch material), and 18 of those

TABLE VII
FINAL MODEL ON THE TWO SEALED SETS, EACH EVALUATED ONCE.

Metric	Kaggle sealed test ($n = 418$)	Web holdout ($n = 407$)
Accuracy	0.837	0.821
95% Wilson CI	0.799–0.870	0.780–0.855
Macro-F1	0.834	0.816
Probe-only accuracy (no zero-shot head)	0.845	0.779
Coverage at threshold 0.5	0.871	0.853
Accuracy on accepted images	0.909	0.891
ECE (15 bins)	0.034	0.050

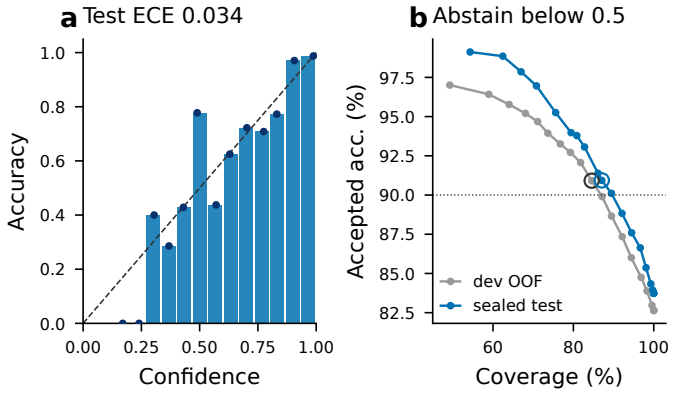


Fig. 5. (a) Reliability diagram on the sealed test ($n = 418$; 15 bins; bar = accuracy, dot = mean confidence; bins with few images are noisy). (b) Coverage vs. accuracy on answered images when abstaining below a confidence threshold (0–0.95, step 0.05) for dev out-of-fold scores (n = 2,512) and the sealed test. Circles: chosen threshold 0.5; dotted: 90% target.

19 are predicted correctly: the test split shares the scrape's contamination.

Calibration and abstention. After temperature scaling the model is well calibrated on unseen data (test ECE 0.034; Fig. 5a). Abstention behaves on the test split much as it did out-of-fold: at threshold 0.5 the model answers 87.1% of test images (out-of-fold 84.6%) and is right on 90.9% of them (out-of-fold 90.9%). Raising the threshold to 0.8 answers 70.8% of test images at 97.0% accuracy.

Export and latency. The image tower and both heads are exported as one ONNX graph (1,637 MB, external-data format) with `logits` and `probs` outputs. In onnxruntime it matches PyTorch to a maximum absolute probability difference of 6.3×10^{-6} with 100% argmax agreement. Batch-1 latency is 892 ms per image on the CPU execution provider and 67 ms with the CUDA provider (16-core desktop CPU, RTX 4060 Ti). OpenCV's DNN importer cannot load the graph (`readNetFromONNX` terminates with a segmentation fault). The demonstrators therefore use onnxruntime for inference. A Python/OpenCV application classifies files, folders, URLs or webcam frames and draws the top-3 labels or *UNSURE*; its webcam and window modes have not yet been tested on a display. Its predictions match the PyTorch reference on 64 dev images (maximum $|\Delta p| = 9.1 \times 10^{-6}$). A Rust/GPUI desktop front end (`apps/gpui-demo`) loads the same ONNX graph through onnxruntime; its Rust port of the preprocessing

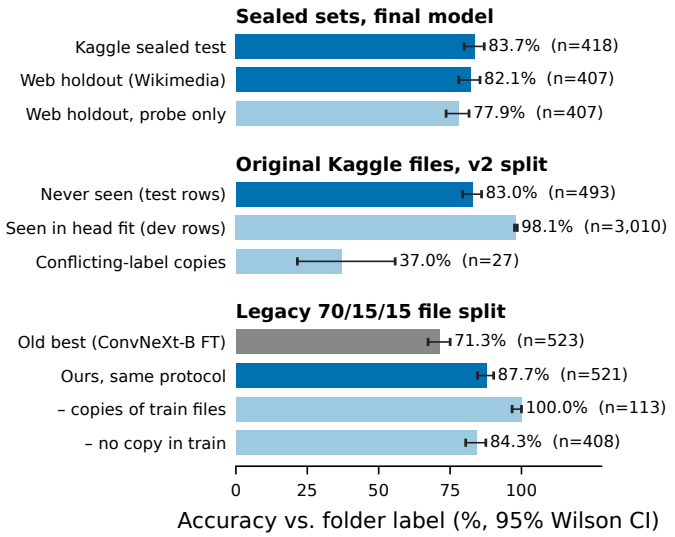


Fig. 6. Accuracy of the final model against the folder label with 95% Wilson intervals. Top: the two sealed sets. Middle: all 3,530 original Kaggle files grouped by the v2 split of their image (dev rows were used to fit the head). Bottom: the earlier 70/15/15 file-level protocol, with our probe refit on legacy-train rows only, against the earlier best fine-tune on the same test files.

agrees with the Python pipeline on 64/64 development images (argmax), and is bit-identical at the input tensor when built against libjpeg-turbo.

IX. GENERALISATION BEYOND THE TEST SPLIT

Fig. 6 collects three further checks of the frozen final model.

Out-of-source web holdout. A second sealed set of 407 Wikimedia Commons photographs (CC/PD licensed, 15–25 per class) was built after the model was frozen. Construction comprised category listing, exact deduplication, and a CLIP ViT-B/32 leakage gate against every training-pool image (cosine $\geq$0.92 dropped; maximum kept similarity 0.903). The nine photos already shown in the demo were excluded, and a strict visual review kept only photos in which one component of the labelled class is dominant. Evaluated once, the model reaches accuracy 0.821 (95% CI 0.780–0.855) and macro-F1 0.816 (Table VII), close to its in-distribution accuracy. Here the zero-shot head pays off. The fused model is correct on 334 images versus 317 for the probe alone (+4.2 points), in line with evidence that retaining a vision–language model's zero-shot classifier preserves robustness to distribution shift [27]. The weakest classes mirror the training contamination: *LED* recall 0.47 (7 errors predicted *diode*), *switch* 0.52 (real toggle, slide and DIP switches predicted *IC chip* 6 times, since the training *switch* class is dominated by game consoles) and *transformer* 0.67 (5 errors to *inductor coil*).

All original Kaggle files. Scored on all 3,530 original files, the model reaches 95.5%. This figure is misleading, because most files are copies of rows used to fit the head (98.1% on 3,010 such files). On the 493 files whose image lies in the sealed test split, which the model has never seen, accuracy is 83.0% (86.5% on the 348 in-scope ones). The 27 files of

conflicting-label duplicate groups score 37.0%: copies of one image carry different folder labels, so they cannot all be right. The legacy test files illustrate the trap. As deployed, the model scores 96.6% on the 523 legacy-test files, but only 71 of them belong to v2 sealed-test rows, and on those it scores 85.9%.

Like-for-like replay of the earlier protocol. To compare with the first iteration on its own terms, we refit the probe only on rows containing a legacy-training file (2,129 rows, $C = 0.1$). We keep the same backbone and zero-shot head and score the legacy-test files. The replay reaches 87.7% ($n = 521$; two conflicting-label files not scored), against 71.3% for the earlier best (a ConvNeXt-B fine-tune) on the same protocol: +16.4 points. Leaked copies of training files score 100% ($n = 113$), and the 408 files with no training copy score 84.3%. The comparison is not fully independent: the backbone and fusion head were chosen on v2 dev folds, which contain many legacy-test images. It nevertheless shows that the gain is not an artefact of the stricter protocol.

X. Limitations and Threats to Validity

One source dataset. All training, CV and sealed-test images come from one Kaggle scrape with shared search-engine biases. The web holdout tests other internet photographs, but bench photographs from a real camera remain untested.

Automated screens. The 27.8% out-of-scope estimate and every in-scope number rest on an unverified CLIP screen with a hand-set threshold. It has false positives (e.g. resistor schematics) and certainly misses some cases. The 69 label issues are likewise model-flagged and unreviewed.

Web-holdout construction. One automated reviewer, looking at thumbnails without a second annotator, kept only clear single-component photos. The holdout is therefore easier than an unbiased web sample. Five classes have only 15 images, and 144 of 984 selected candidates were never downloaded because of rate limits.

Sample size and seeds. The test standard error is ± 1.8 points, and per-class recall rests on 16–25 images. Every configuration was run once per fold with a single seed, so fold SDs (1.0–3.5 points) mix data and optimisation noise.

Unequal tuning. Fine-tunes used one hand-set recipe per family without a per-family search. Probes tuned C by inner grouped CV, and zero-shot used fixed templates. A per-family search could narrow the probe–fine-tune gap for some backbones. The So400m partial fine-tune ran on 3 folds only.

Confounds and selection bias. The schedule fix is confounded with BatchNorm freezing. Fine-tunes early-stop on their scoring fold. The zero-shot fusion weight was suggested by an exploratory dev analysis before being fixed.

Deployment. OpenCV's DNN module cannot run the exported graph, and CPU latency (0.9 s) is too slow for smooth webcam use.

Provenance. The benchmark table was cross-checked against the per-fold run records (maximum difference 4.9×10^{-5}). The leakage simulation, schedule ablation and cleaning ablation summaries were produced by analysis code that is not in the repository; their inputs are.

XI. Conclusion and Future Work

On a small, noisy web-scraped component dataset, the choice of pretraining source outweighs architecture, model size and training compute. A linear probe on frozen SigLIP features beats every fine-tuned CNN and ViT, but fine-tuning is still the better choice for weaker backbones. The data audit matters as much as the model: file-level splits of the scrape leak duplicates, inflate accuracy and compress family differences. Reported component-recognition accuracies should come with a description of how duplicates were handled. The final model meets its 80–85% target on two sealed sets and abstains usefully. Future work: (i) distil the model [47] into a small CNN that OpenCV's DNN module can load and that runs in real time on a CPU; (ii) collect and label target-domain camera photographs; (iii) human relabelling of the 69 label issues and review of the 816 out-of-scope flags, which would make the in-scope subset a verified benchmark; (iv) per-family hyper-parameter search with nested CV and several seeds; and (v) resistor-specific cues such as colour-band reading and higher input resolution.

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